CliniScan: Chest X-ray Lung Abnormality Classification System

Milestone 2: Classification Model Development & Baseline Training

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1. Project Overview & Objective

The objective of this milestone is to develop a **deep learning classification model** capable of identifying abnormalities in **Chest X-ray images**.

The classification system predicts whether an X-ray image is:

- **Normal**
- **Abnormal**

This model acts as the **first stage in the automated medical diagnosis pipeline**, helping filter abnormal cases for further analysis.

2. Dataset Preparation

The dataset used for training contains chest X-ray images from the **VinBigData Chest X-ray dataset**.

Images were preprocessed and organized to train a deep learning classification model.

2.1 Image Preprocessing

The following preprocessing steps were applied:

- Conversion of **DICOM images to PNG format**
- Image resizing to **224 × 224 pixels**
- Normalization of pixel values
- Data augmentation for improving model generalization

These steps ensure consistent input size and improve model performance.

2.2 Dataset Organization

The dataset was organized into **training and validation folders**.

Folder Structure

```
classification_data/
│
├── train/
│   ├── normal
│   └── abnormal
│
└── val/
    ├── normal
    └── abnormal
```

This structure allows the **PyTorch DataLoader** to automatically label images during training.

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3. Classification Model (ResNet18)

The classification model is based on **ResNet18**, a deep convolutional neural network pretrained on **ImageNet**.

Transfer learning was applied by replacing the final fully connected layer to classify chest X-ray images.

Model Architecture

Component	Specification
Base Model	ResNet18
Pretrained Weights	ImageNet
Input Size	224 × 224 RGB
Output Classes	2 (Normal, Abnormal)
Framework	PyTorch

The pretrained backbone allows the model to learn useful image features efficiently.

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4. Training Configuration

The following parameters were used during training.

Parameter	Value
Loss Function	CrossEntropyLoss
Optimizer	Adam
Learning Rate	0.0003
Batch Size	32
Epochs	20
Device	GPU / CPU

The model was trained using **mini-batch gradient descent** to optimize classification accuracy.

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5. Results & Performance

After training, the classification model achieved strong performance on the validation dataset.

Performance Metrics

- **Validation Accuracy:** ~95%
- **Training Loss:** Decreased consistently during training
- **Validation Loss:** Stable across epochs

The results indicate that the model can effectively distinguish between **normal and abnormal chest X-ray images**.

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6. Challenges & Observations

During model development several challenges were observed:

1. Class Imbalance

The dataset contains more **normal images than abnormal images**, which may affect prediction balance.

2. Image Quality Variations

Chest X-ray images vary in brightness and contrast, requiring normalization.

3. Hardware Limitations

Training deep learning models requires **high GPU memory and computation power**.

Despite these challenges, the model successfully learned relevant patterns.

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7. Conclusion

In Milestone 2, a **ResNet18-based classification model** was successfully developed and trained to detect abnormalities in chest X-ray images.

The model achieved **high validation accuracy (~95%)**, demonstrating its capability to distinguish between normal and abnormal scans.

This classification model provides the **foundation for further improvements**, including multi-disease classification and integration with detection models in future milestones.